Vaibhav Satish

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Education

University of California, Irvine: Donald Bren School of Information and Computer Science — Irvine, CA
Bachelor of Science(B.S) Computer Science — *Aug 2024 - Mar 2027*

- **GPA:** 3.873
- **Achievements:** Dean's Honors List (**6/6** consecutive academic quarters)

Work Experience

Software Engineering / Machine Learning Intern — Irvine, CA
Stealth AI Startup — *Jun 2026 - Present*

- Integrated an **OpenAPI API documentation system** utilized by the entire development team, including the **CTO and senior engineers**, by dynamically generating endpoint details from **JSDoc comments**.
- Improved measurement pipeline to process **13** outputs from a **Multilayer Perceptron model**, geometrically calculate **8** length-based measurements, and derive **4** additional circumference measurements using an **anthropometric prior fusing system**.
- Seeded **NeonDB** and **GCP Cloud SQL** databases across **10+** backend endpoints and shipped an optional **brand-filtering** feature enabling brand-specific queries platform-wide.

Software Developer Research Assistant — Irvine, CA
UC Irvine Charlie Dunlop School of Biological Sciences — *Mar 2026 - Present*

- **Scaled the MedTech@UCI cognitive project into a formal lab partnership**, integrating our cognitive game into the MindCycle app for clinical trials.
- Collaborated with the **UCI SLEEP Lab** under **Dr. Katherine Simon**, tracking **4+** core behavioral metrics via **backend logging**.

Software Development Co-Lead — Irvine, CA
MedTech@UCI — *Sep 2024 - Present*

- Co-lead a team of **5** engineers using **Unity** and **C#** to develop a pediatric cognitive analysis app with image-recognition and memory tools, enabling earlier detection of cognitive delays.
- Designed interactive cognitive games using **React** and **JavaScript**, delivering therapeutic interfaces for dementia and Alzheimer's patients.
- Engineered a **React metrics-tracking engine** to capture **15+** core cognitive metrics, providing clinicians with actionable analytical insights.

Software Development Intern & Instructor — Seattle, WA
Tech Academy of Minnesota — *Jun 2025 - Aug 2025*

- Delivered **full-stack** curriculum to **18** students, guiding web development using **Python**, **HTML**, **CSS**, and **JavaScript**.
- Formulated structured **debugging** frameworks and **12** lab exercises, accelerating core engineering and algorithmic problem-solving.
- Translated complex **software engineering paradigms** into digestible concepts, optimizing delivery to boost student project completion rates to **100%**.

Artificial Intelligence Researcher — Remote
Stanford University (via Lumiere Research Program), Georgia Tech & Cornell University (via Inspirit AI Research Program) — *May 2023 - Nov 2023*

- **Developed**, **trained**, and **tuned multi-modal predictive machine learning** models—including **Softmax Regression** and **Deep Neural Networks**—using **PyTorch**, **Scikit-Learn**, **NumPy**, and **Pandas** on a **270,000-row** NFL injury dataset, improving predictive accuracy by **14%** over baseline models.
- Trained **3+ TensorFlow** and **Scikit-Learn** computer-vision models to detect distracted driving, achieving **80-85%** accuracy, and optimized inference pipelines for real-time edge deployment.

Projects

ZOTNest - UC Irvine's 2026 IrvineHacks Hackathon — Feb 2026 - Present
React, Python, FastAPI, Leaflet

- Created a **full-stack property search web app** to streamline housing discovery for **30,000+** UCI students.
- Integrated **Leaflet** to deliver seamless, real-time location mapping and interactive visualization of **30+** local properties.
- Implemented **multi-criteria filtering algorithms** to compute dynamic property recommendations based on custom user vectors.
- Scraped and parsed Google Maps reviews, transforming raw text into qualitative UI insights.

Wordle — Jun 2026 - Present
JavaScript, Python (FastAPI), CSS, React, Next.js

- Deployed a **Wordle** clone with a **FastAPI** backend handling word validation, answer selection, and game-state logic via a **REST API**.
- Built a responsive **React** frontend with real-time keyboard-state tracking, color-coded feedback, and **WordNet**-based word-definition lookups.

Driver Delineation - UC Irvine's 2025 Datathon — Apr 2025
Python, PyTorch, TensorFlow, Scikit-Learn, NumPy

- Trained supervised **ML models** using **PyTorch** and **Scikit-Learn** to predict driver likelihood from high-dimensional datasets.
- Executed **feature engineering** on **40+** impact variables to minimize model overfitting and boost generalizability.
- Designed a multi-layer **Deep Neural Network (DNN)** in **TensorFlow**, tuning hyper-parameters to hit a **90%** validation accuracy.

UCI Themed Geoguessr — Dec 2024 - Present
React, Node.js, TypeScript, C++, Crow API, Google Cloud Platform, Vercel

- Led a **4-person** engineering team from **system architecture ideation** through deployment, managing structured Git workflows.
- Constructed a custom mathematical scoring algorithm and integrated **Leaflet** for sub-second asset rendering and map integration.
- Architected an asynchronous **C++ backend** using **Crow API** and **Google Cloud Platform** to minimize gameplay latency and handle concurrent asset updates.

Technical Skills

Languages/Frameworks: Java, **Python**, **C++**, SQL, **TypeScript**, **JavaScript**, **React**, Node.js, Next.js, **FastAPI**, **REST APIs**, **OpenAPI**, C#
AI/ML Skills: **Machine Learning**, **Supervised/Unsupervised Learning**, **NLP**, **Computer Vision**, **PyTorch**, **TensorFlow**, **Scikit-Learn**, **Pandas**, **NumPy**, **Predictive Modeling**, **Data Analysis**
Databases/Cloud Infrastructure: MySQL, PostgreSQL, **Google Cloud Platform**
Other Skills: HTML, CSS, **Docker**, **Git**, Unity, **Agile**, **CI/CD**, Data Structures, Algorithms, Software Engineering Paradigms